Phenotypic insufficiency of local protein representations: fibre-preserving rearrangements of human C2H2 zinc-finger arrays reorder DNA-recognition residues under an invariant 25-spectrum

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Abstract

A protein representation that depends only on local sequence statistics assigns the same vector to all sequences with the same multiset of k-mers, the *spectral fibre*. Whether this matters depends on the phenotype. For DNA-binding C2H2 zinc-finger proteins it does. Each finger reads about three base pairs through residues at helix positions -1, 2, 3 and 6, and the array reads its target in order. We show that for k at least the length of a finger, the *multiset* of fingers is constant on the fibre, and only their *order* can change. We also give the exact condition for a reordering: the exchanged segments must be bounded by repeated $(k-1)$-mers. The conserved TGEKP linker alone is far too short to bound one at $k = 25$. Among reviewed human proteins, 17 C2H2 zinc-finger proteins have a non-trivial 25-fibre. From 1 178 distinct fibre-mates drawn as random Eulerian trails, 883 (75%) place different recognition residues at some array position. In ZNF91, a KRAB zinc-finger protein that represses SVA and L1 retrotransposons, one fibre-mate changes the recognition residues at 17 of 31 array positions. In ZNF721 one changes 17 of 23. Every window-25 local encoder is constant on these pairs, as verified for all 1 178. Under any finger-additive recognition model, the predicted binding site differs exactly when the ordered recognition residues differ. We formalize this as a phenotypic sufficiency criterion: a representation supports a phenotype if and only if the phenotype is constant on its fibres. We then bound the error of every predictor built on a local encoder. We did not measure binding. The predicted change of specificity rests on the additive model, whose limits we discuss.

Keywords. C2H2 zinc finger, KRAB zinc-finger proteins, DNA-binding specificity, recognition code, spectral fibre, protein representation, sufficiency, ZNF91, ZNF721.

1 Introduction

Protein sequence representations are often built from local statistics: amino acid and k-mer composition, pseudo amino acid composition, and convolutional networks with global pooling. Such a representation is constant on the *spectral fibre* $F_k(S)$ of a sequence, the set of sequences with the same k-mer multiset and the same first $(k-1)$-mer [1, 2, 4]. For 854 reviewed human proteins the 10-mer spectrum does not determine the sequence. C2H2 zinc-finger proteins are 3.0% of the proteome but 40.7% of those proteins [2]. Most of them are KRAB zinc-finger proteins (KZFPs).

That a representation is blind to a set of sequences does not by itself mean that it misses anything a phenotype depends on. Most members of a fibre are not biologically meaningful [5], and a phenotype may be constant on the fibre. The question is therefore phenotype-specific. It needs a phenotype that is known to depend on sequence order and not only on local content.

DNA recognition by tandem C2H2 zinc fingers is such a phenotype. Each finger contacts roughly three base pairs. The fingers bind in antiparallel order, with the N-terminal finger at the $3'$ end of the site, and the site is read as the ordered concatenation of the fingers' sub-sites [10, 14]. KZFPs, with arrays of up to 30 or more fingers, use this to recognize specific transposable-element families [7, 8]. If a fibre-preserving rearrangement reorders the fingers, a local representation cannot distinguish proteins that the recognition code says bind different sites.

This paper makes that argument exact and applies it to the human proteome. Section 2 states the recognition model and its limits. Section 3 proves that fibres preserve the multiset of fingers and gives the condition under which they reorder it. Section 4 computes fibre-mates for all 17 human C2H2 zinc-finger proteins with a non-trivial 25-fibre, with ZNF91 and ZNF721 as case studies. Section 5 formalizes phenotypic sufficiency and bounds what any predictor on a local representation can achieve.

2 The recognition code of tandem C2H2 zinc fingers

A C2H2 zinc finger has the consensus C-$X_{2\text{–}4}$-C-X_{12}-H-$X_{3\text{–}5}$-H. It folds into a $\beta\beta\alpha$ unit around a zinc ion. We number the recognition helix so that the first zinc-coordinating histidine is at position +7; there is no position 0. In the Zif268 structure, residues at helix positions -1, 3 and 6 contact the $3'$, middle and $5'$ bases of a base triplet on the primary strand, and position 2 contacts the complementary strand [10, 6]. Consecutive fingers occupy consecutive triplets in antiparallel orientation, so the site of an array $f_1 \cdots f_n$ reads, from $5'$ to $3'$,

$$Y(S) = g(f_n)\, g(f_{n-1}) \cdots g(f_1), \tag{1}$$

where $g(f)$ is the sub-site (or the position-specific base preferences) of finger f. It depends on the recognition residues $r(f) = (f_{-1}, f_2, f_3, f_6)$ and, to a lesser degree, on other helix residues. We call a recognition model *finger-additive* if it has the form (1) for some map g. Published recognition codes predict g from the helix residues, using structural rules, bacterial one-hybrid data or machine learning [11, 9].

The additive model is an approximation. Neighbouring fingers influence each other through the position-2 contact and through the linker. Not all fingers of a long KZFP array engage DNA at

once, and binding also depends on chromatin [14, 7]. Our conclusions about binding are therefore conclusions about what the additive model predicts. The conclusions about representations do not depend on it.

3 Fibre-preserving rearrangements of finger arrays

Let $S \in \Sigma^L$ and $F_k(S) = \{T : \mathrm{spec}_k(T) = \mathrm{spec}_k(S),\ T[1..k-1] = S[1..k-1]\}$. All members of $F_k(S)$ also share the last $(k-1)$-mer of S [3].

Proposition 1 (the multiset of fingers is fibre-invariant)**.** *For $T \in F_k(S)$ and every word w with $|w| \le k$, w occurs in T as often as in S. In particular, if every finger occurrence has length at most k, which holds for $k \ge 24$ under the consensus above, S and T contain the same multiset of fingers and the same multiset of recognition-residue tuples $r(f)$.*

Proof. Every occurrence of w starts at some position p. If $p \le L - k + 1$, it is the prefix of exactly one k-mer occurrence. Otherwise it lies in the last $(k-1)$-mer. Hence the number of occurrences of w is the number of k-mers in $\mathrm{spec}_k(S)$ with prefix w, plus the number of occurrences of w in the last $(k-1)$-mer. Both are the same for T. A finger occurrence spans 20 to 24 residues. □

So on a fibre of order $k \ge 24$ a local representation cannot lose, gain or alter a finger. It can only fail to see their arrangement. The arrangement changes by the elementary fibre moves: S and T in the same fibre differ by a sequence of exchanges of two adjacent segments, each beginning and ending with an occurrence of the same $(k-1)$-mer [13, 12].

Proposition 2 (when fingers can change places)**.** *Let $S = A\,uXu\,Y\,u\,B$, where u is a word of length $k-1$ and X, Y are arbitrary. Then $T = A\,uY\,u\,X\,u\,B$ lies in $F_k(S)$. Conversely, every $T \in F_k(S)$ arises from S by a finite sequence of such exchanges. In particular, two blocks of fingers can exchange places only if they are bounded by a repeated word of length $k-1$. A linker motif of length ℓ that is identical between fingers bounds such an exchange only for $k \le \ell + 1$.*

Proof. The first part is the standard transposition argument: the k-mers inside uXu and uYu are unchanged, and the k-mers across each boundary read u followed by the first letter of the next block, which is the same multiset in S and T [13]. The converse is the theorem of Ukkonen and Pevzner [13, 12]. The last statement follows because a boundary must be a repeated $(k-1)$-mer. □

The canonical KZFP linker `TGEKP` is 5 residues long and bounds exchanges only for $k \le 6$. At $k = 25$ a boundary must be a 24-residue word that recurs, i.e. two stretches of the array that agree over most of a finger. Such near-identity arises from recent duplication of fingers within an array, the process by which KZFP arrays expand [7].

Proposition 3 (additive models see the order)**.** *Let $T \in F_k(S)$ with $k \ge 24$, and let $r_1, \ldots, r_n$ and $r'_1, \ldots, r'_n$ be the recognition-residue tuples of the finger arrays of S and T in array order. For every finger-additive model whose g depends only on r, $Y(S) = Y(T)$ whenever $r_i = r'_i$ for all i. If g is injective on the tuples that occur, then $Y(S) \ne Y(T)$ whenever $r_i \ne r'_i$ for some i.*

Proof. By Proposition 1 both arrays have n fingers, so $Y(S)$ and $Y(T)$ are concatenations of n sub-sites of equal length, compared position by position. □

Injectivity of g is an idealization. Different fingers can prefer the same triplet, so the numbers below are upper bounds on the pairs for which a given code predicts different sites. They are exact counts of pairs for which the recognition residues are ordered differently.

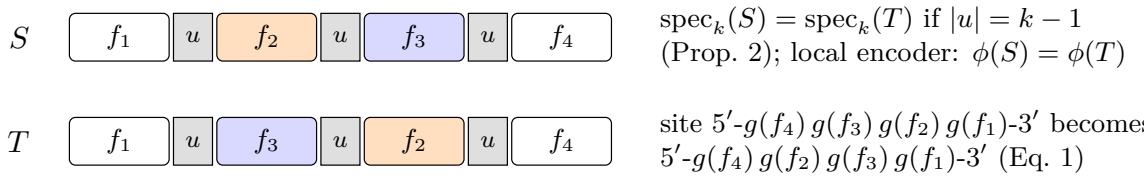


Figure 1: A fibre-preserving exchange of fingers. When the blocks containing f_2 and f_3 are bounded by copies of the same $(k-1)$-mer u (grey), exchanging them preserves the k-spectrum, so every local encoder of order k is unchanged. Under a finger-additive recognition model the predicted site changes whenever $g(f_2) \neq g(f_3)$.

4 Case studies: fibre-mates of human zinc-finger proteins

Method. We took all reviewed human proteins annotated with C2H2-type zinc fingers and with $N_{25} > 1$ in [1]. There are 17. For each, we drew 300 random Eulerian trails of the de Bruijn graph of order 25 starting at the protein's first 24-mer, and kept the distinct sequences $T \neq S$. These draws are not uniform on the fibre. They serve as witnesses, not as a sample. For every mate we verified $\mathrm{spec}_{25}(T) = \mathrm{spec}_{25}(S)$, parsed the fingers of S and T with the consensus motif, checked that the finger multisets agree (Proposition 1), compared the recognition residues position by position along the array, and applied a randomly initialized convolutional encoder of window 25 with mean pooling (float64).

Results. Table 1 lists the proteins. Of 1 178 distinct fibre-mates, 883 (75%) reorder the recognition residues. In every one of them the finger multiset is unchanged and the local encoder output is identical. KLF18 is an instructive exception. Its fibre is large ($\log_2 N_{25} = 12.3$), but its repeats lie outside its three fingers, so none of its 285 mates reorders them. For the binding-site phenotype, a local encoder of order 25 loses nothing on KLF18.

ZNF91. ZNF91 is a primate KZFP with 31 fingers. It binds and represses SVA and L1 retrotransposons, and its array expanded in the primate lineage [8]. Its 25-fibre has $2^{5.0}$ members. In the mate shown in Figure 2, fingers 3–19 are rearranged. Two blocks of the array exchange places across a 24-residue boundary shared by near-identical fingers, and 17 of the 31 array positions carry different recognition residues. Fingers 1–2 and 20–31 are unchanged.

Under (1), the predicted 93-bp binding site changes at 17 of its 31 triplets, while every local encoder of order 25 assigns the wild-type protein and its fibre-mate the exact same representation vector.

ZNF721. ZNF721 has 23 C2H2 fingers and $\log_2 N_{25} = 8.4$. Among the 198 distinct fibre-mates drawn at $k = 25$, 196 (99%) reorder the recognition-residue tuples. In the mate shown in Figure 2b, an internal block permutation across repeated 24-mer boundaries alters the recognition tuples at 17 of the 23 array positions, while preserving the finger multiset and all 25-mer counts.

5 Phenotypic sufficiency and predictor error bounds

We now state the general condition under which an encoder's fibres force prediction errors on a downstream phenotype. Let $\mathcal{X} \subseteq \Sigma^*$ be a domain of sequences, $\phi\colon \mathcal{X} \to \mathcal{Z}$ an encoder with fibres

protein	L	κ	$\log_2 N_{25}$	fingers	distinct mates	new order	max. positions
KLF18	1052	54	12.3	3	285	0	0
ZNF208	1280	95	21.2	36	300	300	33
ZNF254	659	38	1.0	13	1	1	2
ZNF283	679	28	1.0	15	1	1	6
ZNF347	839	35	2.0	17	3	3	6
ZNF594	807	30	1.0	21	1	1	2
ZNF595	648	26	2.0	17	3	3	9
ZNF676	588	38	1.0	14	1	1	2
ZNF709	641	34	4.6	19	23	22	6
ZNF721	911	32	8.4	23	198	196	17
ZNF729	1252	42	9.8	34	246	245	23
ZNF761	746	27	2.0	17	3	3	11
ZNF840P	716	33	1.0	18	1	1	2
ZNF888	718	26	1.0	17	1	1	2
ZNF91	1191	51	5.0	31	31	28	17
ZNF93	620	31	2.6	16	5	5	9
ZNF99	864	49	6.4	21	75	72	9
total					1 178	883	

Table 1: Fibre-mates of order 25 of the 17 human C2H2 zinc-finger proteins with $N_{25} > 1$. "New order": mates in which some array position carries different recognition residues $(-1, 2, 3, 6)$. "Max. positions": the largest number of such array positions in one mate. In all 1 178 mates the finger multiset equals that of S, and the window-25 encoder output is identical to that of S (relative difference below 10^{-12}).

$F_\phi(S) = \{T \in \mathcal{X} : \phi(T) = \phi(S)\}$, and $Y \colon \mathcal{X} \to \mathcal{Y}$ a phenotype map.

Definition 4 (phenotypic sufficiency)**.** An encoder ϕ is *phenotypically sufficient* for Y on $\mathcal{X}$ if Y is constant on every fibre of ϕ, i.e. $\phi(S) = \phi(T)$ implies $Y(S) = Y(T)$ for all $S, T \in \mathcal{X}$.

Theorem 5 (sufficiency criterion and irreducible error)**.** *Let $\phi \colon \mathcal{X} \to \mathcal{Z}$ and $Y \colon \mathcal{X} \to \mathcal{Y}$.*

1. *There exists a predictor $h \colon \mathcal{Z} \to \mathcal{Y}$ with $h(\phi(S)) = Y(S)$ for all $S \in \mathcal{X}$ if and only if ϕ is phenotypically sufficient for Y on $\mathcal{X}$.*
2. *Let d be a metric on $\mathcal{Y}$. For every pair $S, T \in \mathcal{X}$ in the same ϕ-fibre and every predictor $h \colon \mathcal{Z} \to \mathcal{Y}$,*

$$\max\big(d(h(\phi(S)), Y(S)),\ d(h(\phi(T)), Y(T))\big) \ \geq\ \frac{1}{2}\, d(Y(S), Y(T)),$$

and if X is distributed uniformly on $\{S, T\}$, then $\mathbb{E}\, d(h(\phi(X)), Y(X)) \geq \frac{1}{2}\, d(Y(S), Y(T))$.

Proof. (1) If $h \circ \phi = Y$, then $\phi(S) = \phi(T)$ implies $Y(S) = h(\phi(S)) = h(\phi(T)) = Y(T)$. Conversely, if Y is constant on every fibre, setting $h(z) = Y(S)$ for any $S \in \phi^{-1}(z)$ is well defined and satisfies $h(\phi(S)) = Y(S)$. (2) Since $\phi(S) = \phi(T)$, both predictions equal the same point $y = h(\phi(S)) \in \mathcal{Y}$. By the triangle inequality, $d(Y(S), Y(T)) \leq d(Y(S), y) + d(y, Y(T))$, so the sum of the two errors is at least $d(Y(S), Y(T))$ and their maximum (and mean) is at least half of it. □

Combined with Proposition 3 and Section 4, Theorem 5 shows that on the 25-fibres of 16 human C2H2 zinc-finger proteins, no local encoder of order $k \leq 25$ is phenotypically sufficient for

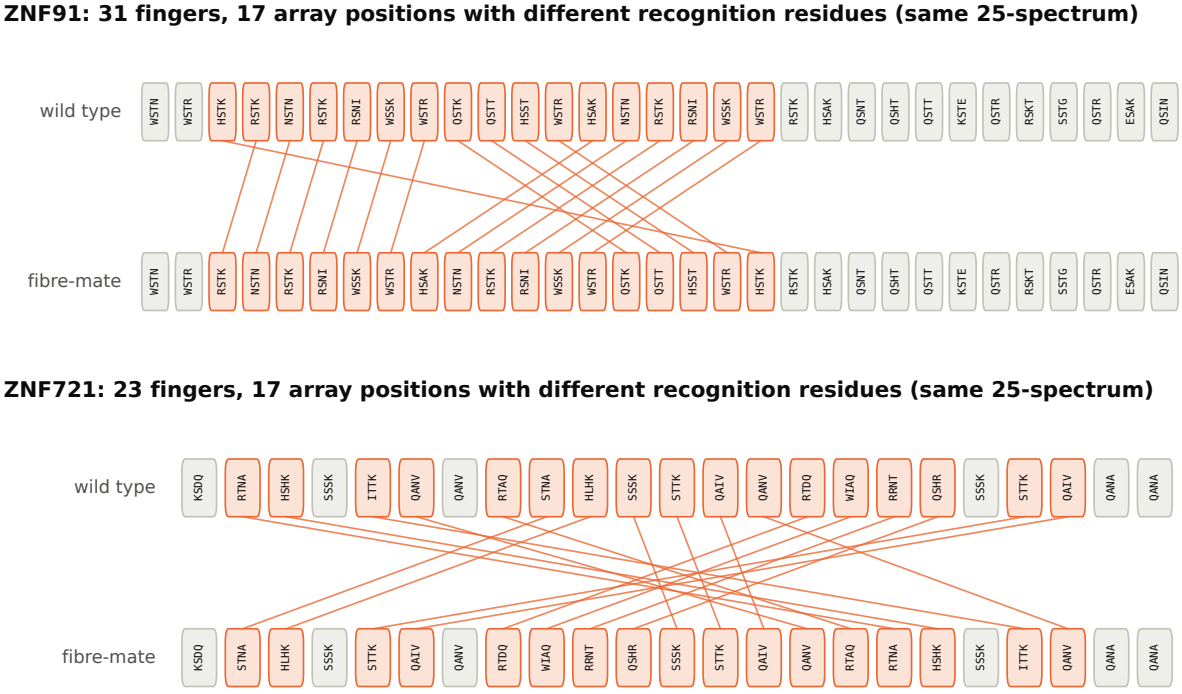


Figure 2: Fibre-preserving finger reorderings at $k = 25$. Array positions of the C2H2 fingers in the wild-type protein S (top) and in a 25-fibre mate $T \in F_{25}(S)$ (bottom) for (a) ZNF91 (31 fingers; 17 positions receive a different recognition quartet) and (b) ZNF721 (23 fingers; 17 positions change). Slanted lines connect fingers that move to a new array position; unmoved fingers are unmarked. In both cases $\mathrm{spec}_{25}(S) = \mathrm{spec}_{25}(T)$ and every window-25 local encoder gives identical outputs on S and T.

any injective finger-additive specificity model: every predictor built on ϕ incurs an error of at least half the motif distance between $Y(S)$ and $Y(T)$ on each reordered fibre pair.

6 Discussion and limitations

The gap between separating a reference set and resolving a sequence becomes a phenotypic blind spot as soon as two conditions meet: the protein's architecture places multiple domain copies between shared $(k-1)$-mer boundaries, and the biological function depends on the order in which those copies sit. Tandem C2H2 zinc-finger arrays satisfy both. At $k = 25$, the fibre preserves the exact collection of fingers (Proposition 1) while permuting blocks of fingers bounded by 24-residue repeats (Proposition 2). For 16 human zinc-finger proteins, including the retrotransposon repressors ZNF91 and ZNF93, such permutations reorder up to 17–33 recognition helices while leaving every order-25 local representation unchanged.

Limitations. First, we did not measure DNA binding experimentally, nor did we run structural docking. The claim that $Y(S) \neq Y(T)$ rests on the finger-additive recognition model (1). While the antiparallel ordering of triplet contacts is established by co-crystal structures and ChIP-exo

footprints [10, 14, 7], only a subset of fingers in a long KZFP array may contact DNA at a given locus, so reordering non-contacting fingers would not alter binding at that locus. Second, the fibre-mates in Section 4 are mathematical Eulerian trails of $G_{25}(S)$, not documented natural alleles; whether unequal crossing-over or ectopic gene conversion produces these specific block permutations in human populations requires long-read polymorphism data. Third, our theorems apply to local encoders. Protein language models with rotary or relative positional encodings are not local [4]; whether their learned representations separate finger-reordered mates is an empirical benchmark question.

Data and code availability

The script `code/finger_order.py` downloads the UniProt human proteome snapshot (or reads it from cache), extracts C2H2 zinc fingers, draws fibre-mates at $k = 25$, verifies 25-spectrum and window-25 CNN invariance, and writes `results/finger_order.json`, `results/finger_order_output.txt` and `results/finger_pairs.tsv`, as well as `figures/fig_ladder.pdf`. Code, results and figures are at `https://github.com/Ruqing1963/zinc-finger-order` and are archived with this paper at `doi:10.5281/zenodo.22998488`.

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References

[1] Z. Chen, R. Chen, *What a local encoder cannot see: spectral fibres of the human proteome and the representation of perturbagens*, Zenodo (2026). `doi:10.5281/zenodo.22950095`

[2] Z. Chen, R. Chen, *Separation without resolution: the spectral fibre geometry of the human proteome and the blind spots of local sequence encoders*, Zenodo (2026). `doi:10.5281/zenodo.22959211`

[3] Z. Chen, R. Chen, *Information loss and quotient geometry of local sequence encoders: representation lattices, fibre entropy and diameter bounds*, Zenodo (2026). `doi:10.5281/zenodo.22960439`

[4] Z. Chen, R. Chen, *What finite-receptive-field encoders cannot distinguish: invariance theorems and architectural boundaries of convolutional sequence models*, Zenodo (2026). `doi:10.5281/zenodo.22963882`

[5] Z. Chen, R. Chen, *Biologically reachable spectral fibres: repeat shuffling, copy ambiguity and the scarcity of cryptic pathogenic missense variants in ClinVar*, Zenodo (2026).

[6] M. Elrod-Erickson, M. A. Rould, L. Nekludova, C. O. Pabo, *Zif268 protein–DNA complex refined at 1.6 Å: a model system for understanding zinc finger–DNA interactions*, Structure **4** (1996) 1171–1180. `doi:10.1016/S0969-2126(96)00125-6`

[7] M. Imbeault, P.-Y. Helleboid, D. Trono, *KRAB zinc-finger proteins contribute to the evolution of gene regulatory networks*, Nature **543** (2017) 550–554. doi:10.1038/nature21683

[8] F. M. J. Jacobs et al., *An evolutionary arms race between KRAB zinc-finger genes ZNF91/93 and SVA/L1 retrotransposons*, Nature **516** (2014) 242–245. doi:10.1038/nature13760

[9] H. S. Najafabadi et al., *C2H2 zinc finger proteins greatly expand the human regulatory lexicon*, Nat. Biotechnol. **33** (2015) 555–562. doi:10.1038/nbt.3128

[10] N. P. Pavletich, C. O. Pabo, *Zinc finger–DNA recognition: crystal structure of a Zif268–DNA complex at 2.1 Å*, Science **252** (1991) 809–817. doi:10.1126/science.2028256

[11] A. V. Persikov, M. Singh, *De novo prediction of DNA-binding specificities for C2H2 zinc finger proteins*, Nucleic Acids Res. **42** (2014) 97–108. doi:10.1093/nar/gkt890

[12] P. A. Pevzner, *DNA physical mapping and alternating Eulerian cycles in colored graphs*, Algorithmica **13** (1995) 77–105. doi:10.1007/bf01188582

[13] E. Ukkonen, *Approximate string-matching with q-grams and maximal matches*, Theoret. Comput. Sci. **92** (1992) 191–211. doi:10.1016/0304-3975(92)90143-4

[14] S. A. Wolfe, L. Nekludova, C. O. Pabo, *DNA recognition by* Cys_2His_2 *zinc finger proteins*, Annu. Rev. Biophys. Biomol. Struct. **29** (2000) 183–212. doi:10.1146/annurev.biophys.29.1.183